

# The `lalign` Package

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## Abstract

This is a collection of macros that I developed while writing my dissertation. They are mostly focused on displaying, naming and referencing mathematical programming problems. The `lalign` environment macro formats math programs for legibility in both code and display. The named paragraph and sub-equation macros are useful in any document which defines and references multiple problems, algorithms or multi-line environments.

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## 1 Installation

Copy `lpalign.sty` to the same directory as your main `.tex` file and add the command `\usepackage{lpalign}` to your preamble. The `amsmath` and `expl3` packages are required but will also be added by the above command. I recommend using these macros in conjunction with the `hyperref` package so that custom tags/references are clickable.

## 2 The `lpalign` Environment

`lpalign` is an environment macro built on the `alignat` environment from the `amsmath` package. It formats mathematical programs so that they are legible in code as well as in the final pdf. Its primary function is to automatically format and position the ‘sense’ (maximize or minimize) and ‘subject to’ terms within a gutter to the left of the `alignat` body. These terms are aligned with the objective and first constraint respectively.

The `lpalign` environment takes two arguments—`<sense>` and `<objective>`—along with several optional keys:

- `vars` – subscript on the ‘sense’ term. Commonly used to state variables and ranges.
- `vars-style` – font style for the `vars` value. Defaults to `\scriptstyle`.
- `st` – text used for the ‘subject to’ term. Defaults to `s.t..`
- `objective-sep` – controls the spacing between objective function and the first constraint. The default value of `0em` can be changed during package load by the `lp_objective-sep` option.  
e.g. `\usepackage[lp_objective-sep=XXem]{lpalign}`.
- `qualifier-sep` – controls the spacing between constraint body and the qualifier (for-all) term. The default value of `2em` can be changed during package load by the `lp_qualifier-sep` option.  
e.g. `\usepackage[lp_qualifier-sep=XXem]{lpalign}`.
- `gutter-sep` – controls the spacing between ‘sense’/‘subject to’ terms and the constraint body. The default value of `0.75em` can be changed during package load by the `lp_gutter-sep` option.  
e.g. `\usepackage[lp_gutter-sep=XXem]{lpalign}`.
- `sense-shift` – adds a vertical shift to the ‘sense’ term.
- `st-shift` – adds a vertical shift to the ‘subject to’ term.

The `lpalign*` environment is identical without printing equation reference tags unless explicitly called for by the `\tag` command.

```

\begin{lpalign}{Maximize}{\mathbf{c}^\top\mathbf{x}}
  A\mathbf{x} \leq \mathbf{b} \quad \backslash
  \mathbf{x} \geq \mathbf{0}
\end{lpalign}

```

↓

$$\text{Maximize } \mathbf{c}^\top \mathbf{x} \quad (1)$$

$$\text{s.t. } A\mathbf{x} \leq \mathbf{b} \quad (2)$$

$$\mathbf{x} \geq \mathbf{0} \quad (3)$$

Just like in vanilla alignat, ampersands & control alignment; the constraint rows will be aligned at the ampersand.

```

&Ax+s = b+0 \quad Ax+s &= b+0 \quad Ax+s = b+0& \quad \backslash
&x,s \geq 0 \quad x,s &\geq 0 \quad x,s \geq 0&

```

↓

|                               |                               |                               |
|-------------------------------|-------------------------------|-------------------------------|
| $\max \quad c^\top x$         | $\max \quad c^\top x$         | $\max \quad c^\top x$         |
| $\text{s.t. } Ax + s = b + 0$ | $\text{s.t. } Ax + s = b + 0$ | $\text{s.t. } Ax + s = b + 0$ |
| $x, s \geq 0$                 | $x, s \geq 0$                 | $x, s \geq 0$                 |

## Qualified Constraints

Qualified constraints are also possible. A double-ampersand && will properly space and align a ‘for-all’ (∀) statement.

```

\begin{lpalign*}{Maximize}{\sum_{j=1}^m c_j x_j}
  \mathbf{a}_i^\top \mathbf{x} \leq b_i
  \quad \quad \quad \&\& \forall i \in [n] \quad \backslash
  x_i \geq 0 \quad \quad \quad \&\& \forall i \in \{1,2,\dots,n\}

```

↓

$$\begin{aligned} &\text{Maximize} && \sum_{j=1}^m c_j x_j \\ &\text{s.t.} && \mathbf{a}_i^\top \mathbf{x} \leq b_i \quad \forall i \in [n] \\ &&& x_i \geq 0 \quad \forall i \in \{1, 2, \dots, n\} \end{aligned}$$

## Variable Ranges

Subscripts can be placed under the ‘sense’ term via the vars option.

```
\begin{lpalign*}[vars=\mathbf{x}\in\mathbb{R}_+]
  {Maximize}{c_1x_1 + \dots + c_mx_m}
  A\mathbf{x} &\leq \mathbf{b}
\end{lpalign*}
```

↓

$$\begin{array}{ll} \text{Maximize} & c_1x_1 + \dots + c_mx_m \\ \mathbf{x} \in \mathbb{R}_+ & \\ \text{s.t.} & A\mathbf{x} \leq \mathbf{b} \end{array}$$

## Whitespace

The `sep` options can be used to modify the whitespace between elements. Their default values are mutable at package load.

```
\begin{lpalign}
[
objective-sep=2em,
qualifier-sep=4em,
gutter-sep=4em,
]
{Maximize}{\mathbf{c}^{\top}\mathbf{x}}
A\mathbf{x} &\leq \mathbf{b} \\
\mathbf{x} &\geq \mathbf{0}
\end{lpalign}
```

↓

$$\begin{array}{ll} \text{Maximize} & \sum_{j=1}^m c_j x_j \end{array} \quad (4)$$

$$\text{s.t.} \quad \mathbf{a}_i^\top \mathbf{x} \leq b_i \quad \forall i \in \llbracket n \rrbracket \quad (5)$$

$$x_i \geq 0 \quad \forall i \in \{1, 2, \dots, n\} \quad (6)$$

Remember that space can be added between individual rows (or lines of text) via an optional argument on `\\`.

## Multiline Objectives

The `objective function` is wrapped in an `aligned` environment from `amsmath`. This means that very long objectives can easily be expressed in multiple lines rather than overflowing the margins or creating awkward spacing.

```

\begin{lpalign}{Maximize}
{
  c_1&x_1+c_2x_2+c_3x_3+c_4x_4 \\\
  &+c_5x_5+\dots+c_mx_m
}
\begin{aligned}
a_{i1}&x_1+a_{i2}x_2+a_{i3}x_3 \\\
&+a_{i4}x_4+\dots+a_{im}x_nm
\end{aligned}
\end{aligned} &\leq b_i \quad \forall i \in [n] \\\
\mathbf{x} &\geq \mathbf{0}
\end{lpalign}

```

↓

$$\begin{array}{ll} \text{Maximize} & c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4 \\ & + c_5x_5 + \dots + c_mx_m \end{array} \quad (7)$$

$$\text{s.t.} \quad \begin{array}{l} a_{i1}x_1 + a_{i2}x_2 + a_{i3}x_3 \\ + a_{i4}x_4 + \dots + a_{im}x_nm \end{array} \leq b_i \quad \forall i \in [n] \quad (8)$$

$$\mathbf{x} \geq \mathbf{0} \quad (9)$$

Multiline `constraints` can be displayed by nesting their own `aligned` environments in the constraint cells. Tags and qualifiers will be placed correctly.

The `'sense'` and `'subject to'` terms can be freely shifted up or down

```

\begin{lpalign}
[ sense-shift=0.5\baselineskip,
  st-shift=0.5\baselineskip ]
...
...
\end{lpalign}

```

↓

$$\begin{array}{ll} \text{Maximize} & c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4 \\ & + c_5x_5 + \dots + c_mx_m \end{array} \quad (10)$$

$$\text{s.t.} \quad \begin{array}{l} a_1x_1 + a_2x_2 + a_3x_3 \\ + a_4x_4 + \dots + a_mx_m \end{array} \leq b_i \quad (11)$$

### 3 Named Paragraphs

The `namedpara` command displays a paragraph marker that takes an abbreviation, which is useful for referencing specific problems or conventions.

```
\namedpara{Named Paragraph}[NPg]\label{npara1} \\
Paragraph \ref{npara1} is a named paragraph.
```

↓

**Named Paragraph (NPg)**

Paragraph **NPg** is a named paragraph.

Currently supported optional keys include

- abv** – abbreviation used when referencing. A single argument with no = sign is also interpreted as this abbreviation.
- indent** – indent distance. The default value of 2em can be changed during package load by the `namedpara_indent` option.  
e.g. `\usepackage[namedpara_indent=XXem]{lalign}`.
- before** – arbitrary code placed between the indent and the title.
- after** – arbitrary code placed after the abbreviation.
- flush** – centers or right-flushes the paragraph title via `center` or `right`.

```
\namedpara{Indented}[abv=IND, indent=6em]
```

```
\namedpara{Surrounded}[abv=SUR, before=$>>>$, after=$<<<$]
```

```
\namedpara{Rightflushed}[abv=RIGHT, flush=right]
```

↓

**Indented (IND)**

**>>>Surrounded (SUR)<<<**

**Rightflushed (RIGHT)**

The compatibility command `paratitle` is left untouched from previous versions.

## 4 Named Sub-equations

The `namedsubeqs` environment uses a given name as the first index in any enclosed equation reference tag. A second index is appended as in a traditional `subequations` environment.

```

\begin{namedsubeqs}{EQ}
\begin{align}
A\mathbf{x} &= \mathbf{b} & \label{eq:1} & \\
C\mathbf{y} &= \mathbf{d} & \label{eq:2} & \\
\end{align}
\end{namedsubeqs}

```

Equations `\ref{eq:1}` and `\ref{eq:2}` are equations.

↓

$$Ax = b \quad (\text{EQ.a})$$

$$Cy = d \quad (\text{EQ.b})$$

Equations [EQ.a](#) and [EQ.b](#) are equations.

An optional argument allows you to change the style of the sub-tag. The currently supported keys are `style` for the second-index formatter (default `\alph`) and `punct` for the separator between the main tag and the sub-tag (default `.`).

```

\begin{namedsubeqs}{EQ}[style=\roman]

\begin{namedsubeqs}{EQ}[style=\arabic]

```

↓

|          |         |          |        |
|----------|---------|----------|--------|
| $Ax = b$ | (EQ.i)  | $Ax = b$ | (EQ.1) |
| $Cy = d$ | (EQ.ii) | $Cy = d$ | (EQ.2) |

Fonts can be changed using the appropriate switch commands.

```

\begin{namedsubeqs}{EQ}[style=\itshape\roman]

\begin{namedsubeqs}{EQ}[style=\scshape]

```

↓

|          |         |          |        |
|----------|---------|----------|--------|
| $Ax = b$ | (EQ.i)  | $Ax = b$ | (EQ.A) |
| $Cy = d$ | (EQ.ii) | $Cy = d$ | (EQ.B) |

The `punct` argument controls the separator between the main tag and the sub-tag.

```

\begin{namedsubeqs}{EQ}[style=\scshape\roman, punct=:]

\begin{namedsubeqs}{EQ}[style=\bfseries\arabic, punct=$\sim$]

```

↓

|                                                                         |                                                                        |
|-------------------------------------------------------------------------|------------------------------------------------------------------------|
| $A\mathbf{x} = \mathbf{b}$ (EQ:I)<br>$C\mathbf{y} = \mathbf{d}$ (EQ:II) | $A\mathbf{x} = \mathbf{b}$ (EQ~1)<br>$C\mathbf{y} = \mathbf{d}$ (EQ~2) |
|-------------------------------------------------------------------------|------------------------------------------------------------------------|

A single optional argument which does not contain an = sign is interpreted as the `style` argument.

```

\begin{namedsubeqs}{EQ}[\sffamily\Alph]

\begin{namedsubeqs}{EQ}[\bfseries\arabic]

```

↓

|                                                                        |                                                                        |
|------------------------------------------------------------------------|------------------------------------------------------------------------|
| $A\mathbf{x} = \mathbf{b}$ (EQ.A)<br>$C\mathbf{y} = \mathbf{d}$ (EQ.B) | $A\mathbf{x} = \mathbf{b}$ (EQ.A)<br>$C\mathbf{y} = \mathbf{d}$ (EQ.B) |
|------------------------------------------------------------------------|------------------------------------------------------------------------|

## 5 Sub-References

Also available are `\lpsubref` and `\lpsubeqref`, which work like `\ref` and `\eqref` but reference only the secondary, sub-index part of a tag inside either `namedsubeqs` or a traditional `subequations` environment from the `amsmath` package.

```

\begin{namedsubeqs}{SQ}[style=\itshape\roman]\label{nq}
\begin{align}
A\mathbf{x} = \mathbf{b} \quad \label{nq:1}\\
C\mathbf{y} = \mathbf{d} \quad \label{nq:2}
\end{align}
\end{namedsubeqs}

```

Just like `\ref{nq:1}` and `\eqref{nq:2}`, we can reference an equation's sub-tag `\lpsubref{nq:1}` and `\lpsubeqref{nq:2}`. The environment itself may also be referenced `\ref{nq}`.

↓

$$\begin{array}{ll}
 A\mathbf{x} = \mathbf{b} & \text{(SQ.i)} \\
 C\mathbf{y} = \mathbf{d} & \text{(SQ.ii)}
 \end{array}$$

Just like `SQ.i` and `(SQ.ii)`, we can reference an equation's sub-tag `i` and `(ii)`. The environment itself may also be referenced `SQ`.

`\lpsubref` and `\lpsubeqref` work with ordinary `\label` inside both `namedsubeqs` and traditional `subequations` environments. Other environments are not officially supported.

```

\begin{subequations}\label{sq}
\renewcommand{\theequation}
{\theparentequation\textbar\arabic{equation}}
\begin{align}
A\mathbf{x} = \mathbf{b} \quad \label{sq:1}\\
C\mathbf{y} = \mathbf{d} \quad \label{sq:2}
\end{align}
\end{subequations}

```

We can also reference `\ref{sq}`, `\ref{sq:1}` and `\eqref{sq:2}`, or sub-reference `\lpsubref{sq:1}` and `\lpsubeqref{sq:2}` in a `\textsf{subequations}` environment.

↓

$$\begin{array}{ll}
 A\mathbf{x} = \mathbf{b} & (12|1) \\
 C\mathbf{y} = \mathbf{d} & (12|2)
 \end{array}$$

We can also reference `12`, `12|1` and `(12|2)`, or sub-reference `1` and `(2)` in a `subequations` environment.

Supported punctuation currently includes the period (`.`), colon (`:`), semicolon (`;`), comma (`,`), hyphen (`-`), forward slash (`/`), and `\textbar` (`|`). Other punctuation

has not been tested.

## 6 Combining Macros

Each of these macros may be combined for a nicely named, labeled, formatted and tagged mathematical programming problem.

```

\paratitle{Non-Linear Program}[NLP]\label{para:NLP}
\begin{namedsubeqs}{NLP}
\begin{lpalign}{Minimize}{f(\mathbf{x}) \quad \label{NLP.0}}
g_i(\mathbf{x}) \ \&\ge\ 0 \ \&\&\forall i \in I \ \label{NLP.1} \ \&\&
h_j(\mathbf{x}) \ \&\le\ 0 \ \&\&\forall j \in J \ \label{NLP.2} \ \&\&
\mathbf{x} \ \&\ge\ 0 \quad \label{NLP.3}
\end{lpalign}
\end{namedsubeqs}

```

Problem \eqref{para:NLP} is a nonlinear program.  
Equation \eqref{NLP.0} is the objective function.  
Equations (\ref{NLP.1}--\ref{NLP.3}) are the constraints.  
Non-negativity is enforced by constraint \lpsubeqref{NLP.3}.



### Non-Linear Program (NLP)

$$\begin{array}{ll} \text{Minimize} & f(\mathbf{x}) \quad (\text{NLP.i}) \\ \text{s.t.} & g_i(\mathbf{x}) \geq 0 \quad \forall i \in I \quad (\text{NLP.ii}) \\ & h_j(\mathbf{x}) = 0 \quad \forall j \in J \quad (\text{NLP.iii}) \\ & \mathbf{x} \geq 0 \quad (\text{NLP.iv}) \end{array}$$

Problem (NLP) is a nonlinear program. Equation (NLP.i) is the objective function. Equations (NLP.ii-iv) are the constraints. Non-negativity is enforced by constraint (iv).

## 7 Spacing Macros

## Quick Spacing

Also included are font-relative versions of the common spacing commands `\quad` and `\qquad` and a few more for good measure:

- |                  |  |  |                |  |  |
|------------------|--|--|----------------|--|--|
| • \hskip = 0.5em |  |  | • \HSkip = 3em |  |  |
| • \Hskip = 1em   |  |  | • \HSKIP = 4em |  |  |
| • \HSkip = 2em   |  |  |                |  |  |

Negative versions of the above spacing macros:

- `\htrg = -0.5em`
- `\Htrg = -1em`
- `\HTrg = -2em`
- `\HTRg = -3em`
- `\HTRG = -4em`

Vertical versions of the above horizontal spacing macros:

- `\vskp = 0.5em`
- `\Vskp = 1em`
- `\VSkp = 2em`
- `\VSKp = 3em`
- `\VSKP = 4em`
- `\vtrg = -0.5em`
- `\Vtrg = -1em`
- `\VTrg = -2em`
- `\VTRg = -3em`
- `\VTRG = -4em`

These are, I would say, less reliable than using `\vspace` or `\vskip` correctly, but I find them more convenient.

Some horizontal spacing between these `\HSPK` words `\`

Some vertical spacing between these `\VSKp` lines `\`

Some negative-horizontal spacing between these `\HTrg` words `\`

Some negative-vertical space between these `\Vtrg` lines, so that they overlap.

↓

Some horizontal spacing between these words

Some vertical spacing between these

lines

Some negative-horizontal spacing between these words

Some negative-vertical space between these lines, so that they overlap.

## Summation Spacing

Some macros for correctly spacing display-style summation notation when the lower index set is overfull. This is done by placing the subscript in a zero-width box and should not be used in inline math.

- $\backslash\text{SUM} = \sum$
- $\backslash\text{CUP} = \bigcup$
- $\backslash\text{VEE} = \bigvee$
- $\backslash\text{PROD} = \prod$
- $\backslash\text{CAP} = \bigcap$
- $\backslash\text{WEDGE} = \bigwedge$

These each take two arguments

- M. The lower index set
- O. The upper index

This spacing is a personal preference; I think it makes things more consistent and legible. The below example is comically exaggerated, please use `\substack` to illuminate any *really* long indexing.

$$\begin{aligned}
 X &= \text{\sum_{i\in way too much indexing} x_i} \\
 X &= \text{\SUM{i\in way too much indexing}x_i} \\
 X &= \text{\SUM{\substack{i\in way to \\ much indexing}}x_i} \\
 X &= \text{\SUM{i=1,300,000}[1,300,001]x_i} \\
 &\downarrow \\
 X &= \sum_{i \in \text{waytoomuchindexing}} x_i & X &= \sum_{i \in \text{waytoomuchindexing}} x_i \\
 X &= \sum_{\substack{i \in \text{wayto} \\ \text{muchindexing}}} x_i & X &= \sum_{i=1,300,000}^{1,300,001} x_i
 \end{aligned}$$

## Equation Spacing

A few binary relations with extra space on either side. I like these for aligned inequality constraints.

- $\backslash\text{EQ} =$
- $\backslash\text{IN} \in$
- $\backslash\text{NEQ} \neq$
- $\backslash\text{NIN} \notin$
- $\backslash\text{LS} <$
- $\backslash\text{LE} \leq$
- $\backslash\text{SUB} \subset$
- $\backslash\text{SEQ} \subseteq$
- $\backslash\text{GS} >$
- $\backslash\text{GE} \geq$
- $\backslash\text{SUP} \supset$
- $\backslash\text{SPQ} \supseteq$

$$\begin{aligned}
 &\begin{array}{l} Ax+s \text{ \&eq b } \\ x,s \text{ \&ge 0} \end{array} & \begin{array}{l} Ax+s \text{ \&EQ b } \\ x,s \text{ \&GE 0} \end{array} \\
 &\downarrow \\
 &\begin{array}{l} \max \quad c^\top x \\ \text{s.t. } Ax + s = b + 0 \\ x, s \geq 0 \end{array} & \begin{array}{l} \max \quad c^\top x \\ \text{s.t. } Ax + s = b + 0 \\ x, s \geq 0 \end{array}
 \end{aligned}$$

## 8 Known Bugs

- ☐ Very long objective functions may overrun the page. For now, use an embedded `aligned` environment to display a long function on multiple lines.
- ☒ Spacing macros `\quad` and `\qqquad` overwrote default, absolute behavior with font-relative spacing. Fixed by renaming macros to `\Hskip` and `\HSkip`; v1.1.
- ☒ The package depended on `makecell` and `vcell` without using them. Fixed by removing those unused dependencies; v1.2.
- ☒ The `namedsubeqs` environment assigned `\currentlabel` without switching to `\makeatletter` scope. Fixed internally; v1.2.
- ☒ The `\lpsubref` and `\lpsubeqref` macros failed outright when `hyperref` was not loaded. Fixed by falling back to plain text references when `hyperref` is absent; v1.2.
- ☒ When `hyperref` was loaded, `\sublabel` could produce duplicate equation destinations or occasionally link to the wrong tag. Fixed by assigning dedicated `hyperref`-facing equation identifiers and by removing a redundant equation-counter decrement at the end of `namedsubeqs`; test branch after v1.2.
- ☒ Ordinary `\label` now works for `\lpsubref` and `\lpsubeqref` inside both `namedsubeqs` and traditional `subequations`. Fixed on the test branch after v1.2.
- ☒ The public commands `\subref` and `\subeqref` conflicted with packages such as `subcaption`. Fixed by renaming the package commands to `\lpsubref` and `\lpsubeqref`; v1.3.
- ☒ The objective function did not contribute correctly to the width of the environment. Fixed in v1.4.

## 9 Version History

- v1.0 Dec. 20, 2025. Initial distribution. Includes `lalign` and `namedsubeqs` environments along with macros for named paragraphs and spacing.
- v1.1 Jan. 30, 2026. Bug fix. Renamed quick spacing macros to maintain compatibility with standard  $\TeX$ . Added negative, vertical and negative-vertical versions. Added summation spacing macros. Added clause alignment arguments to the `lalign` environment.
- v1.2 Jun. 25, 2026. Bug fix. Removed unused package dependencies, fixed the internal `\currentlabel` handling in `namedsubeqs`, added a no-`hyperref` fallback for `\lpsubref` and `\lpsubeqref`, and added `\lpsubref`/`\lpsubeqref` support for the traditional `subequations` environment from `amsmath`.
- v1.3 Jun. 26, 2026. Feature and compatibility update. Added `\namedpara` with key-value options, namespaced the sub-equation reference commands as `\lpsubref` and `\lpsubeqref`, and hardened parent-label handling for both `namedsubeqs` and traditional `subequations`.
- v1.4 Jun. 28, 2026. Layout and defaults update. Reworked `lalign` so the

objective contributes correctly to the environment width, ordinary rows no longer need trailing `&&` placeholders, and package-load defaults are now available for `objective-sep`, `qualifier-sep` and `gutter-sep`.